

COURSE MATERIAL

SUBJECT	BASIC CIVIL AND MECHANICAL ENGINEERING (ME23AES101)
UNIT	III
COURSE	B.TECH
DEPARTMENT	Civil Engineering
SEMESTER	I-I or I-II
PREPARED BY (Faculty Name/s)	Mrs. G. Gnana Prasanna, Assistant Professor
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1. Course Objectives

The objectives of this course are to

- Get familiarized with basic Construction Materials; the scope and importance of Civil Engineering specializations.
- Introduce the preliminary concepts of Structural and Geotechnical Engineering.
- Acquire preliminary knowledge on Surveying and Transportation Engineering.
- Get familiarized with the importance of Water Resources and Environmental Engineering.

2. Prerequisites

Nil

3. Syllabus

UNIT III

Surveying & Transportation Engineering: Objectives and Principles of Surveying- Instruments used in Surveying-Importance of Transportation in Nation's economic development- Modes of Transportation- Types of Highway Pavements- Flexible Pavements and Rigid Pavements -Traffic signals and signs.

Water Resources and Environmental Engineering: Introduction to Hydrology– hydrological cycle- Sources of water- Quality of water- Specifications- Rainwater Harvesting- Water Storage and Conveyance Structures- Dams and Reservoirs-types and components.

4. Course outcomes

After successful completion of the course, students will be able to:

1. Understand various sub-divisions of Civil Engineering and to appreciate their role in ensuring better society and the basic characteristics of Construction Materials.
2. Gain knowledge regarding Structural and Geotechnical Engineering.
3. Explain the concepts of surveying and Transportation Engineering, Water Resources and Environmental Engineering.

5. CO-PO / PSO Mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	3											2	2
CO2	3	3											2	2
CO3	3	3											2	2

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Objectives and Principles of Surveying Instruments used in Surveying	T1, R1
2		Importance of Transportation in Nation's economic development- Modes of Transportation	T1, R1
3		Types of Highway Pavements- Flexible Pavements and Rigid Pavements -Traffic signals and signs	T1, R1
4	2	Introduction to Hydrology– hydrological cycle	T1, R2
5		Sources of water- Quality of water- Specifications	T1, R1
6		Rainwater Harvesting	T1, R1
7	3	Water Storage and Conveyance Structures	T1, R1
8		Dams and Reservoirs-types and components.	T1, R1

7. Activity Based Learning

1. Five-minute speech on any important Roads and Dams
2. Physical tests on water for testing its quality.

8. Lecture Notes

3.1 INTRODUCTION

The specializations like surveying, transportation engineering, water resources engineering and environmental engineering are playing pivotal role in the development of the society.

3.2 SURVEYING

- Surveying is the art of determining the relative positions of points above or below the surface of the earth.
- This can be achieved by the measurement of distances, directions and elevations.
- Surveying is typically used in construction, land development, and mapping to create accurate representations of land, buildings, or other structures.

3.2.1 Objectives of Surveying

- The main object of any survey is the preparation of plan or map showing all the features of the area under consideration.
- A plan may be defined as a projection of the ground and the features upon it on a horizontal plane.
- If the scale is large, then it is called a plan.

- If the scale is small, then it is called a map.
- A map is a visual representation of a large area, while a plan is a detailed drawing of a small area.

Example: A map of India, a plan of a building.

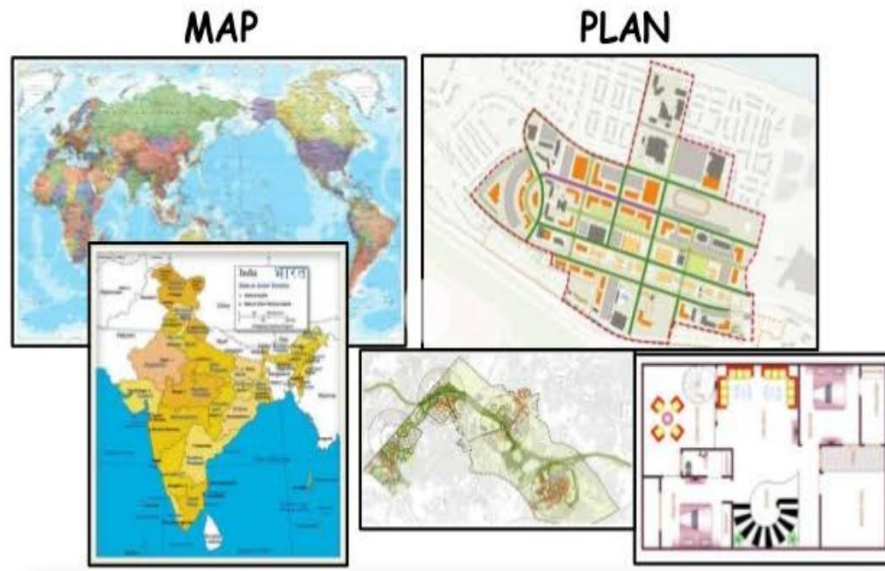


Fig.1: Difference between Map and Plan

3.2.2 Principles of Surveying

The basic principles of surveying are

1. To work from whole to the part
2. To locate a new point by at least two measurements from fixed reference points.

1. To work from whole to the part

- Surveyors start by measuring large, overarching points or boundaries (the "whole") and then break it down into smaller sections (the "part").
- The whole area is first enclosed by mean of stations and main survey lines.
- The area is then divided into a number of parts by forming well-conditioned triangle.
- This principle ensures that any errors in measurement can be traced back to the larger framework, making it easier to detect mistakes and correct them.

2. To locate a new point by at least two measurements from fixed reference points

- In surveying the relative positions of points are located by measurements from at least two points of reference whose positions are known.

▪ The position of new points from the two known points can be fixed or located by taking

- Linear measurements
- Angular measurements
- Both linear and angular measurements
- For example, from known positions of P and Q, the position of a new point R is fixed or located by any of the following method.

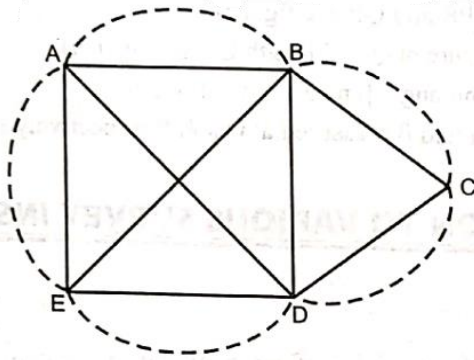


Fig. 2a work from whole to the part

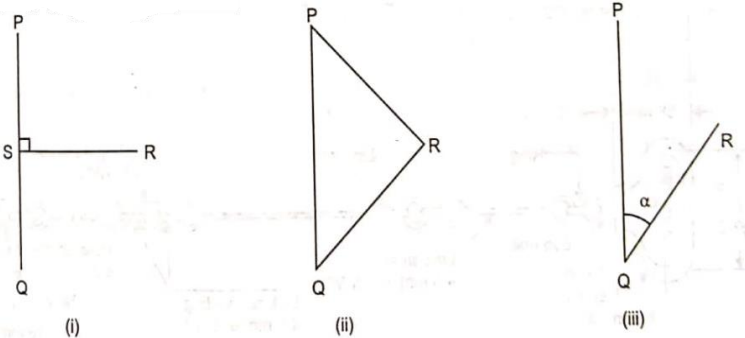


Fig. 2b Location of a new point

3.2.3 INSTRUMENTS USED IN SURVEYING

Surveying involves the use of various instruments to measure distances, angles, and heights.

The common instruments used for surveying are

- Distance Measurement Instruments
- Angle Measurement Instruments
- Levelling Instruments
- Other Instruments

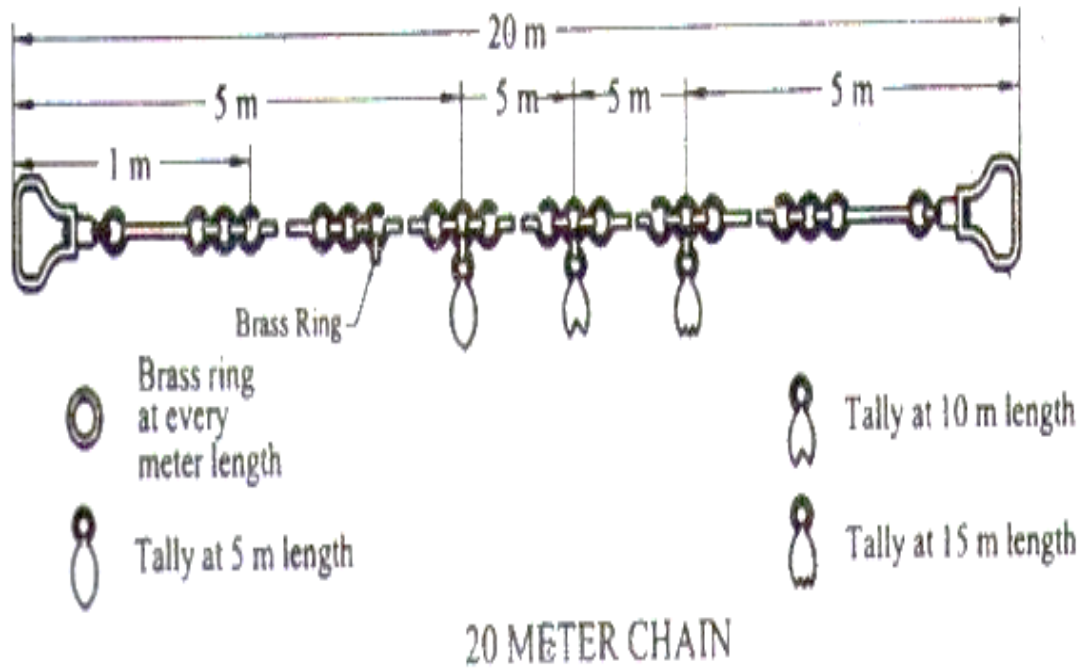
1) Distance Measurement Instruments

Distance measurement instruments in surveying are used to measure the distance between two points.

The common distance measurement instruments used in surveying are

a) Chain or Survey Chain

- Used for measuring distances in older methods, especially in land surveys.
- The chain is generally composed of 100 or 150 links.
- The links are formed by pieces of galvanized mild steel wire 4 mm in diameter.
- The chains are available in lengths of 20 m and 30 m.



b) Measuring Tape

- A measuring tape is used for more accurate measurements of length.
- It typically comes in lengths of 10 to 100 meters or more.
- It is used for measuring smaller areas.
- Linen tape, metallic tape, steel tape and invar tape are the different types of tapes generally used in surveying.



Cloth or Linen Tape



Steel Tape



Invar Tape

c) Electronic Distance Measurement Instruments (EDM Instruments)

- EDM devices use electromagnetic waves (often infrared or laser) to measure distances by calculating the time it takes for a signal to travel from the instrument to the target and back.
- These instruments are highly accurate and commonly used in combination with total stations for modern surveys.

2) Angle Measurement Instruments

In surveying, measuring angles is crucial for determining the direction or position of points relative to each other.

Various instruments are used to measure horizontal, vertical, and inclination angles with high accuracy.

The common angle measurement instruments used in surveying are

Theodolite

Transit Theodolite

Total Station

Compass

a) Theodolite

Theodolite measures both horizontal and vertical angles with great precision.

Theodolite is used in most surveying projects, such as land surveys, construction, and mapping.

Theodolites are highly accurate and come with graduated scales to measure angles down to seconds of arc.

b) Transit Theodolite

Transit theodolite is very similar to a theodolite but designed for more rugged field use, measuring both horizontal and vertical angles.

Transit theodolite is commonly used in engineering, forestry, and construction surveys.

The transit can be easier to handle in rough terrain.

It features a telescope that can be rotated 360 degrees horizontally and adjusted vertically to measure angles.

c) Total Station

- A modern surveying instrument that combines an electronic theodolite and an EDM.
- It can measure both angles and distances accurately, providing 3D spatial data.
- Total stations are widely used for land surveys and construction layouts.



Total Station+Prism+Pole+Tripod

A set of Total Station



d) Compass

- Compass measures horizontal angles relative to magnetic north.
- Compass is used in basic or preliminary surveys.
- Compass surveys are quick and inexpensive but less precise than other methods.
- It is commonly used for small-scale surveys.



3) Levelling Instruments

- Levelling instruments in surveying are tools used to measure height differences between points on the Earth's surface and to establish a horizontal plane of reference.
- They are crucial for determining elevation or vertical position during surveying tasks like construction, road design, and mapping.

The different levelling instruments used in surveying are

- Dumpy level
- Auto level
- Tilting level
- Laser level

a) Dumpy level

- A basic, optical instrument with a telescope and a spirit level tube. Used for measuring horizontal levels and elevation differences between points.



- Most commonly used for general construction work.

b) Auto level

- Similar to the dumpy level, but with a self-leveling mechanism.
- More efficient and accurate for quick and precise leveling tasks.



c) Tilting level

A type of leveling instrument that uses a tilting mechanism to establish a level line of sight.



d) Laser level

A type of leveling instrument that uses a laser beam to establish a level line of sight.



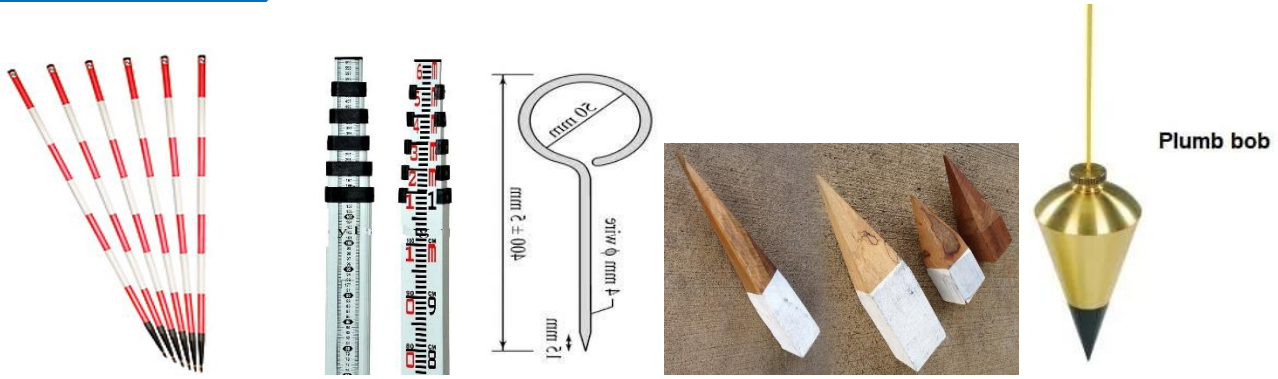
4) Other Instruments

The following instruments are used in various surveying techniques

- a) Ranging rods
- b) Levelling staff
- c) Arrows
- d) Pegs
- e) Plumb bob

a) Ranging rods

- Ranging rods are surveying instruments used to mark the position of survey stations.
- They can be made of wood, steel, or aluminum.
- Ranging rods are circular in section and generally 2 m to 3 m in length.
- These rods are painted with alternate bands of red and white color or black and white.



b) Leveling Staff

A graduated staff used to measure heights.

c) Arrows

- Surveying arrows, also known as chaining pins or survey stakes, are used to mark boundaries and important points in surveying.
- They are made of durable materials like steel.
- Arrows help to precisely locate points in the ground, when we are surveying huge areas.

d) Pegs

- Survey pegs, also known as surveyors' stakes or pins, are used to mark points, lines, and boundaries on the ground.
- They are a crucial tool for surveying.
- The pegs are made of hard wood of 25 mm X 25 mm section, 150 mm long with one end tapered.
- These are driven in to the ground to mark the instrument station.

e) Plumb Bob

- In surveying, a plumb bob is used to determine verticality, and mark positions on the ground.
- It's also used to center surveying instruments like theodolites, prismatic compass and plane table.

3.3 TRANSPORTATION ENGINEERING

3.3.1 Importance of Transportation in Nation's Economic Development

- Transportation plays a vital role in a nation's economic development.
- Transportation is a critical component of the economic development of any country.
- It is the backbone of trade, commerce, and the movement of people and goods.
- It plays a vital role in the movement of goods, people, and services from one location to another, making it possible for businesses to access markets and consumers to access products and services.

The key reasons why transportation is vital for economic growth are

1) Facilitates Trade and Commerce

a) Connects markets

Transportation connects producers to markets, enabling the exchange of goods and services.

b) Promotes international trade

Efficient transportation networks facilitate international trade, increasing exports and economic growth.

2) Supports Economic Growth

a) Creates jobs

- Transportation infrastructure projects create jobs and stimulate economic growth.
- The transportation sector itself provides millions of jobs in industries like logistics, shipping, aviation, construction, and maintenance.

b) Attracts investments

Well-developed transportation networks attract investments, promoting economic growth and development.

3) Boosts Industrial Growth

Access to reliable transportation allows industries to expand their production capacities by ensuring that raw materials are sourced timely and products are delivered efficiently.

4) Supports Regional Development

- Transportation infrastructure, such as roads, bridges, and rail systems, can help connect remote or underdeveloped regions to larger cities or markets.
- This can reduce regional inequalities by providing access to education, healthcare, and employment opportunities, stimulating local economies.

5) Improves Quality of Life

a) Enhances mobility

Transportation enables people to access employment, education, healthcare, and other essential services.

b) Increases access to markets

Transportation improves access to markets, enabling people to purchase goods and services.

6) Improves Productivity and Efficiency

- With faster and more efficient transportation options, businesses can reduce production and delivery times, improving overall productivity.

- In turn, this leads to cost savings and greater competitiveness in global markets.

7) Fosters Tourism

- Transportation systems enable the movement of tourists, which is a significant contributor to many national economies.
- Good transportation infrastructure makes a country more accessible and attractive to visitors, generating revenue in the hospitality, service, and retail sectors.

8) Strengthens National Integration

- Transportation plays a key role in uniting different regions of a country.
- It allows goods, services, and people to move freely across borders, facilitating cultural exchange, reducing regional disparities, and promoting national unity.

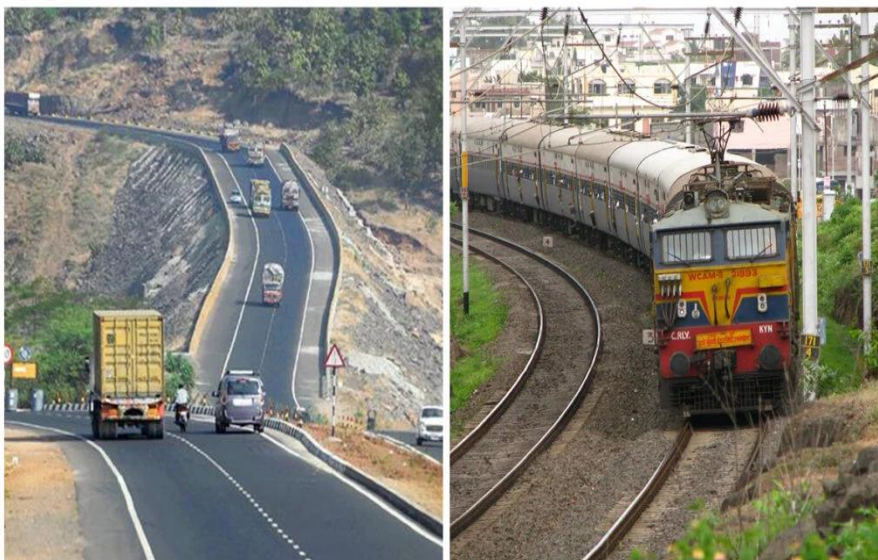
3.3.2 Modes of Transportation

There are three modes of transportation, they are

- 1) Land ways
- 2) Water ways
- 3) Air ways

1) Land ways

- Land transport covers all land-based transportation systems that provide for the movement of people, goods and services.
- Land transport plays a vital role in linking communities to each other.
- Land transport is a key factor in urban planning.
- It consists of two types; they are roadways and railways.



Road Ways

- The transportation by road is the only mode which could give maximum service to one and all.
- This is the most common and flexible mode of transportation, involving vehicles such as cars, buses, trucks, and motorcycles traveling on roads or highways.
- Road transport is ideal for short to medium distances and is essential for both passenger and goods transport.

Advantages

- Flexibility in routes
- Door-to-door service is possible
- Lower cost for short distances

Disadvantages

- Traffic congestion
- Weather-dependent
- Limited capacity
- Pollution

Railways

- Railway is another important long way transport facility.
- Rail transport involves the use of trains running on rail tracks.
- It is highly efficient for transporting large quantities of goods, such as coal, minerals, and manufactured products, as well as passengers over long distances.

Advantages

- High capacity
- Energy-efficient,
- Suitable for heavy and bulk goods
- Safer for long-distance travel

Disadvantages

- Limited to fixed routes
- High infrastructure costs
- Slower compared to air travel

2) Water ways

- Transportation by water is the slowest among all modes of transportation.



- Waterways transportation is the movement of people and cargo by boat, ship, barge, or sailboat on rivers, canals, lakes, and oceans.
- It is a vital and economical way to move large quantities of heavy items over long distances.

Advantages

- High capacity
- Cost-effective for bulk cargo
- Suitable for international trade

Disadvantages

- Slow mode when compared to other modes
- Dependent on weather and sea conditions
- Requires ports and infrastructure.



3) Airways

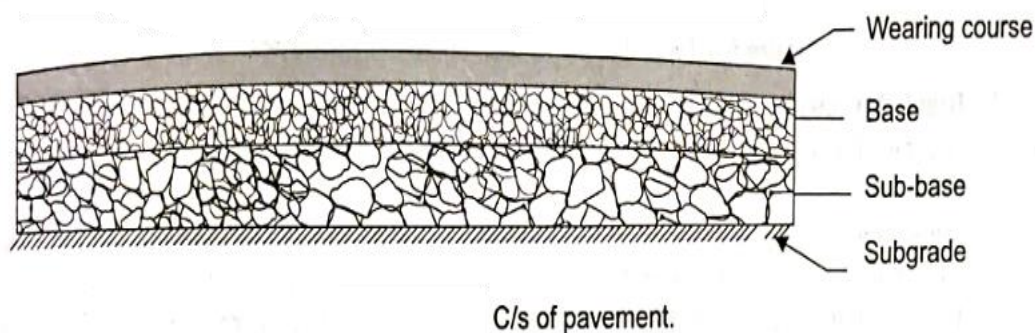
- The transportation by air is the fastest among all modes of transportation.
- This mode uses aircraft, such as airplanes and helicopters, to transport passengers and goods through the air.
- Air travel also provides more comfort and saves transportation time for the passengers.



3.3.3 PAVEMENT

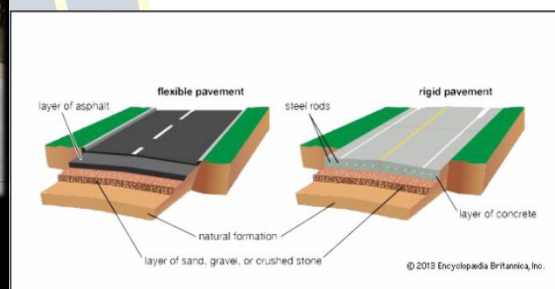
- Pavement structure is the combination of subbase, base and wearing or surface course placed on a subgrade to support the traffic load and distribute it to the road bed.

- The purpose of the pavement is to provide a functional surface for the safe operation of a vehicle.



TYPES OF HIGHWAY PAVEMENTS

- Highway pavements are designed to provide a durable, smooth, and safe surface for vehicles.
- Majorly there are two types of pavements, flexible and rigid pavements each suited for different environmental conditions, traffic loads, and budget considerations.



1) Flexible Pavement

- Flexible pavements are designed to flex under traffic loads, distributing the weight through multiple layers.
- Flexible pavements are typically composed of multiple layers of asphalt concrete, with each layer having a different thickness and composition.
- In flexible pavements, wheel loads are transferred by grain-to-grain contact of the aggregate through the granular structure.
- The flexible pavement has less flexural strength.
- Flexible pavement requires periodic maintenance such as resurfacing and patching to prevent deterioration.

Advantages of flexible pavement

- Cost-effective initial construction.

- Quick and easy to repair.
- Provides a smooth riding surface.
- Adaptable to varying traffic loads.

Disadvantages of flexible pavement

- Shorter lifespan than rigid pavement
- Requires frequent maintenance compared to rigid pavements.
- The maintenance cost is high.
- Susceptible to oil stains and damage from other chemicals.
- Edges are weak and therefore require curb structures or edging

2) Rigid Pavement

- Rigid pavement is a type of road pavement that is constructed using concrete and has a high flexural strength.
- Rigid pavement spreads the load over a wider area due to its stiffness.
- Rigid pavements exhibit low deflection under traffic loads, which reduces the risk of cracking.

Advantages of Rigid Pavement

- Longer lifespan.
- Maintenance costs are low.
- Rigid pavement can be installed on low- and high-quality soils.
- More resistant to weather conditions and fuel/oil spills.
- Rigid pavement can handle heavy traffic loads efficiently.

Flexible pavement	Rigid Pavement
➤ Load is transfer by grain to grain.	➤ Load transfer is not exists.
➤ Low strength.	➤ High strength.
➤ Low life Span.	➤ Life span is more.
➤ High maintenance cost.	➤ Low Maintenance cost.
➤ Rolling of surface is needed.	➤ Rolling of surface is not needed.
➤ Damaged by oils and certain chemicals.	➤ No damage by oils and greases.

3.3.4 Traffic Signals and Signs

1. The typical colors used in traffic signals are

1) Red

2) Yellow or Amber

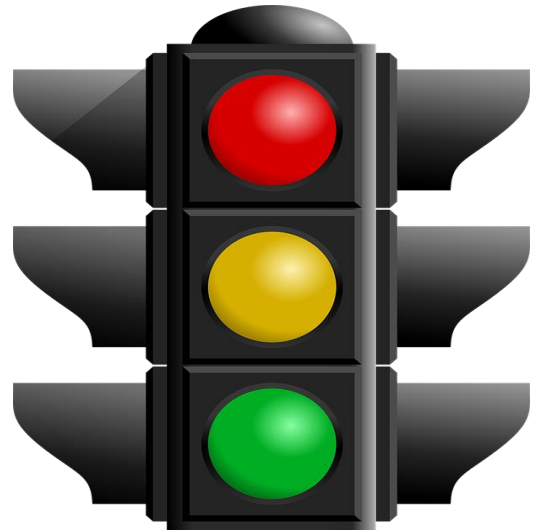
3) Green

1) Red

- It signifies stop.
- When the light is red, vehicles and pedestrians must come to a complete stop and wait until the light changes.

2) Yellow or Amber

- This color indicates caution.
- It usually comes on before the red light, signaling that the light is about to change to red.
- It tells drivers and pedestrians to prepare to stop.



3) Green

- **Green means go.**
- When the light is green, vehicles and pedestrians are allowed to proceed through the intersection.

Advantages of Traffic Signals

Traffic signals offer several advantages that contribute to safer and more efficient traffic management

- They provide orderly movement of traffic.
- They increase the traffic handling capacity of most of the intersections at grade.
- They reduce accidents.
- Pedestrians can cross the roads safely at the signalized intersection.

Disadvantages of Traffic Signals

While traffic signals offer numerous advantages, they also come with certain disadvantages, they are

- Improperly timed or poorly coordinated traffic signals can contribute to congestion, especially during peak traffic hours.
- While pedestrian signals ensure safety, long wait times for pedestrians to cross intersections can be a drawback.
- Failure of the signal due to electric power failure or any other defect may cause confusion to the road users.

2. TRAFFIC SIGNS

- Traffic Signs are the signs erected at the side of roads to provide information to road users.
- They are also known as road signs.
- Traffic signs provide valuable information to drivers and other road users.
- They help to ensure road safety and smooth traffic flow by conveying important information quickly and clearly.
- They are an essential part of traffic management.
- The traffic signs should be backed by law in order to make them useful and effective.

Purpose

- To help drivers understand road rules
- To ensure road discipline
- To provide road instructions
- To warn drivers about potential dangers
- To regulate driver behavior
- To enhance road safety

Types of Traffic Signs

Traffic signs have been divided in to three categories according to Indian Motor Vehicle Act.

They are

- 1) Guide or Informatory signs
- 2) Regulatory or Mandatory signs
- 3) Warning or cautionary signs

1) Guide or Informatory Signs

- These signs are used to guide and give useful information to the road users.
- Guide signs provide information about routes, destinations, and services (e.g., highway exits, mile markers, hospital signs).
- Guide signs are usually rectangular in shape.
- Guide signs have blue or green backgrounds.
- Use symbols and text to show directions, distances, or facilities.
- Guide signs make travel easier, safer, and more pleasant.

INFORMATIVE SIGNS



Facility information signs



****no through road**; this sign is used at the entrance to a road from where there is no exit

Direction and place identification signs



2) Regulatory or Mandatory Signs

- Regulatory or mandatory traffic signs are signs that tell road users what they must or must not do.
- They are legally enforceable, meaning not following them can lead to fines or penalties.
- Regulatory or mandatory signs are usually circular in shape with a red border and white background or blue background.

Some of the regulatory signs are;

- Certain speed limit;
- No entry;
- No parking;
- Prohibit of use of horns;
- U turn prohibited;
- Left turn or right turn prohibited; etc.



3) Warning or Cautionary Signs

- Warning or cautionary traffic signs are signs used to alert drivers and other road users about potential dangers or changes in the road ahead.
- These signs are crucial for road safety and help drivers take necessary precautions.
- Their main purpose is to give advance notice so people can slow down, stay alert, or take precautions to avoid accidents.
- Warning or cautionary signs are typically triangular in shape having a red border with a white or yellow background.

As per the I.R.C standardization, a warning sign is in the form of an equilateral triangle with side of **450mm** supported on post at a height of **2750mm** above the ground level up to the base of triangle.

Some of the typical warning signs are;

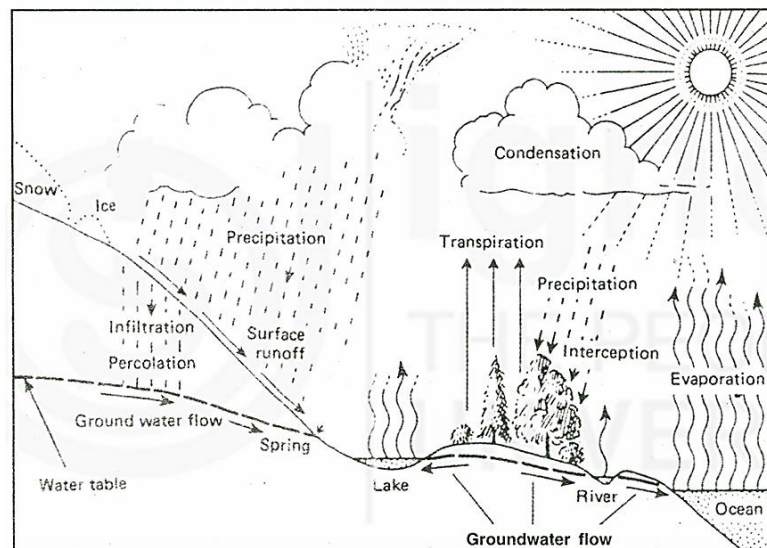
- Cross roads;
- Narrow bridge;
- Dead end or side road;
- School;
- Steep hill;
- Hair pin bend;
- Zig zag turns;
- Men at work; etc



3.4 HYDROLOGY

Hydrology is the science dealing with all aspects of water including its occurrence, movement, circulation and distribution of water of the earth and its surface.

3.4.1 Hydrological Cycle



3.4.2 Sources of water

- Rainwater
- Surface water (rivers, lakes)
- Groundwater (wells, springs)
- Desalinated water
- Recycled water

3.4.3: Quality of water- Specifications

Drinking Standards

Fluoride - 1.5 mg/l

Arsenic – 0.05 mg/l

Iron – 1 mg/l

Nitrates – 100 mg/l

Chlorides – 200 mg/l

pH – 6.5 to 8.5

Calcium – 75 to 200 mg/l

Sulphate – 200 to 400 mg/l

Alkalinity – 200 to 600 mg/l

Turbidity – 5 to 10 NTU

Simple treatment methods

Turbidity – Cloth filtration

Colour – Carbon filtering

Odour – boiling

Bacterial impurities – boiling

Fluorides – Alumina Technology

Ammonia – boiling

Iron – Oxidation & Settling

Hardness – boiling & filtration

Chlorides – boiling & settling

3.4.4 Rain water harvesting

- Rainwater harvesting is a technology used for collecting and storing rainwater from rooftops, the land surface or rock catchments using simple techniques such as jars and pots as well as more complex techniques such as underground check dams.
- Rainwater harvesting is the accumulation and deposition of rainwater for reuse on-site, rather than allowing it to run off.

Uses of rain water

- Recharge underground water
- Gardening
- Livestock
- Drinking purpose
- Irrigation purpose

Advantages:

- Rainwater harvesting technologies are simple to install and operate.

- Local people can be easily trained to implement such technologies, and construction materials are also readily available.
- Maintenance costs are almost negligible.
- Water collected from roof catchments usually is of acceptable quality for domestic purposes.

Methods of harvesting rainwater

Broadly there are two ways harvesting rainwater

- Surface runoff harvesting
- Roof top rainwater harvesting

Surface runoff harvesting: -

In urban area rainwater flows away as surface runoff. This runoff could be caught and used for recharging aquifers by adopting appropriate methods.

Rooftop rainwater harvesting: -

It is a system of catching rainwater where it falls. In rooftop harvesting, the roof becomes the catchments, and the rainwater is collected from the roof of the house/building. This method is less expensive and very effective.

Components of the roof top rainwater harvesting system

- Catchment area
- Transportation
- First flush
- Filtration system
- Storage system
- Delivery system

Catchment area

- The surface that receives rainfall directly is the catchment of rainwater harvesting system. It may be terrace, courtyard, or paved or unpaved open ground.



Transportation

- Rainwater from rooftop should be carried through down take water pipes or drains to storage/harvesting system.



First Flush

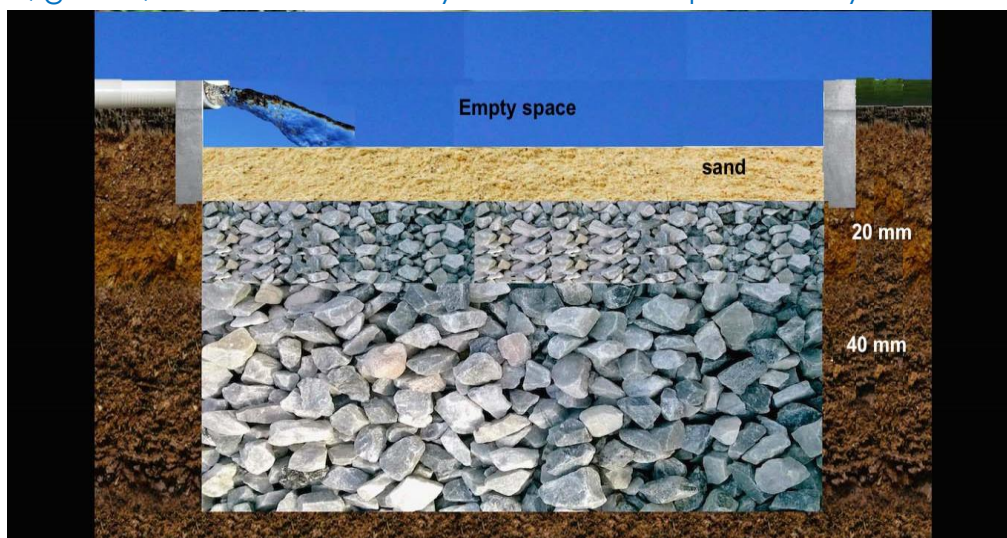
- First flush is a device used to flush off the water received in first shower.
- The first shower of rains needs to be flushed-off to avoid contaminating storable/rechargeable water by the probable contaminants of the atmosphere and the catchment roof.
- It will also help in cleaning of silt and other material deposited on roof during dry seasons. Provisions of first rain separator should be made at outlet of each drainpipe.

Filtration system

- Filters are used for treatment of water to effectively remove turbidity, color and microorganisms.
- After first flushing of rainfall, water should pass through filters.
- There are different types of filters in practice, but basic function is to purify water.

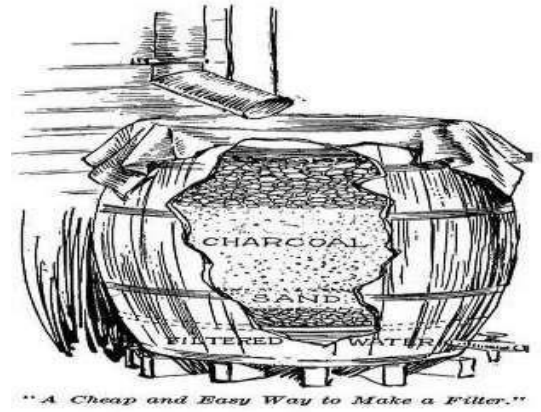
Sand Gravel Filter

These are commonly used filters, constructed by brick masonry and filled by pebbles, gravel, and sand. Each layer should be separated by wire mesh.



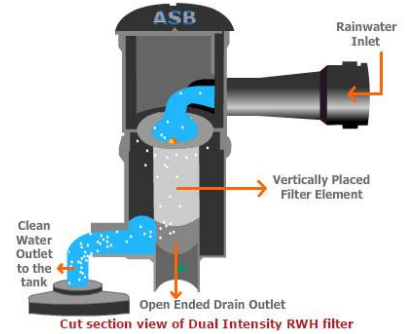
Charcoal filter

Charcoal filter can be made in-situ or in a drum. Pebbles, gravel, sand and charcoal should be filled in the drum or chamber. Each layer should be separated by wire mesh. Thin layer of charcoal is used to absorb odour if any.



PVC-Pipe filter

- This filter can be made by PVC pipe of **1 to 1.20 m** length.
- These are provided with the vertical filter elements to remove the dirt present in water.

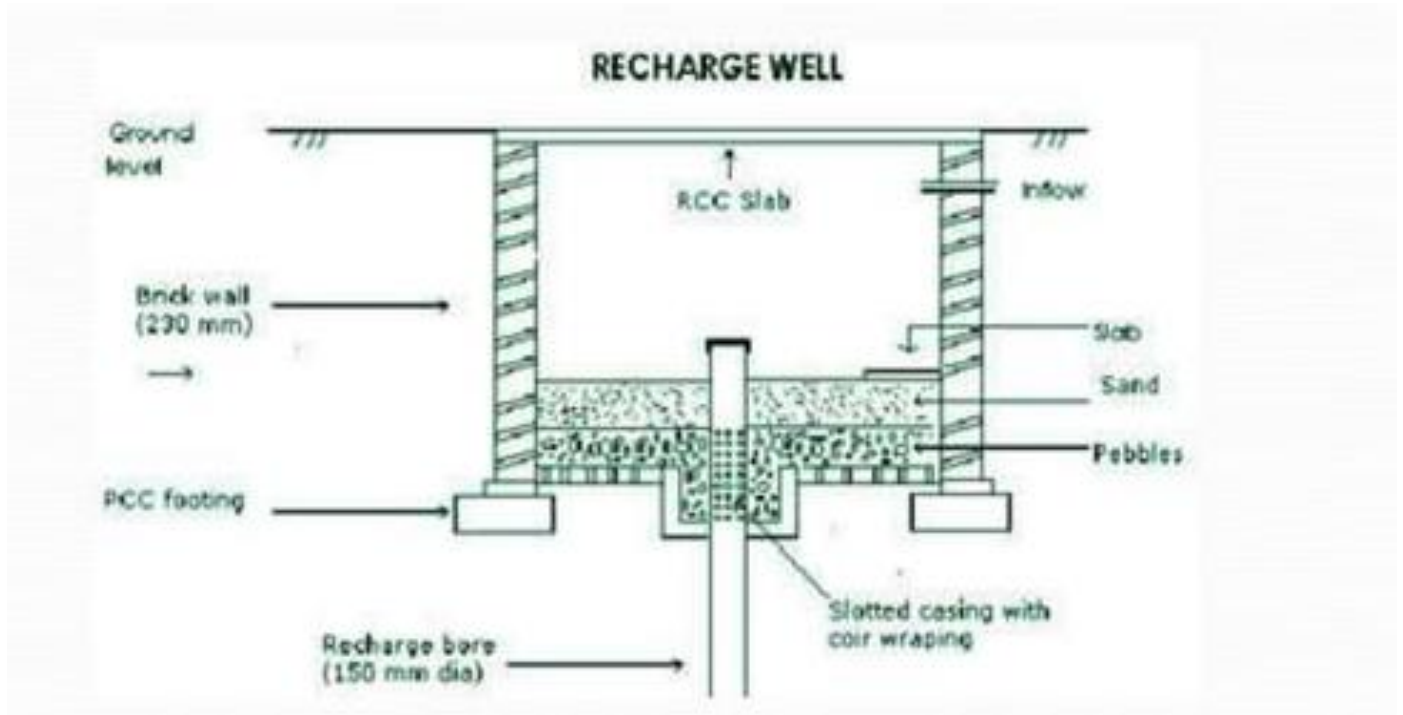


Storage system and Delivery system

All collected rain water are stored in tank or barrels. It is a system to deliver water for uses. There is use of pumps to take out water from tank and deliver for many purposes. Water is delivered by pipes.



Recharging of Dug wells

**Importance of rain water harvesting**

Rainwater is a precious resource due to increases in demand from our ever-growing population. Changes in rainfall patterns in the regions have seen both drought conditions and flooding

- Rainwater harvesting can provide around 50 per cent of a family's water needs.
- This not only saves water, but saves money and reduces our impact on the environment.
- To inculcate a culture of water conservation
- To reduce soil erosion

Examples

- In the state of Tamil Nadu, rainwater harvesting was made compulsory for every building to avoid ground water depletion.
- In Rajasthan, rainwater harvesting has traditionally been practiced by the people of the Thar Desert.
- At present, in Pune (in Maharashtra), rainwater harvesting is compulsory for any new society to be registered.

3.5 Water storage and conveyance structures

Water-conveying structures can refer to various systems designed to transport water efficiently. Some common examples include:

1. **Aqueducts:** These are engineered channels or structures used to transport water over long distances, often using gravity. They have been used historically by civilizations like the Romans and still exist in modified forms today.

2. Pipelines: These are modern structures made of various materials like metal, plastic, or concrete, used to transport water from one place to another. Pipelines are commonly used for transporting water over long distances or for specific purposes like irrigation, industrial use, or delivering potable water.
3. Canals: Man-made waterways designed for navigation, irrigation, or water conveyance. Canals can be large-scale, like the Panama Canal, or smaller, for local water distribution or transportation purposes.
4. Dams and Reservoirs: While not strictly conveyance structures, they are essential for storing water and regulating its flow. Dams can control water flow, ensuring a steady supply downstream, while reservoirs store water for various purposes such as irrigation, drinking water, or electricity generation.
5. Water Towers and Tanks: These are localized structures used to store and distribute water within communities. Water towers use gravity to create pressure in water distribution systems, ensuring a steady supply of water to homes and buildings.

3.6 Dams and Reservoirs-types and components

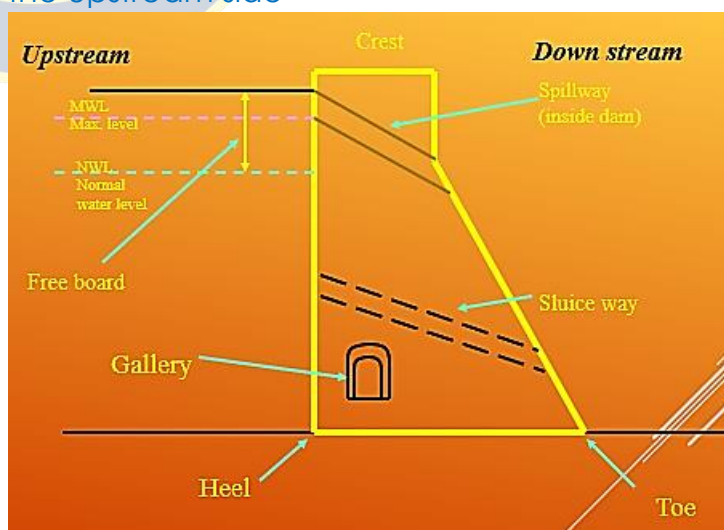
Dam: Dam is a solid barrier constructed at a suitable location across a river valley to store flowing water.

Storage of water is utilized for following objectives:

- Hydropower
- Irrigation
- Water for domestic consumption
- Drought and flood control
- For navigational facilities
- Other additional utilization is to develop fisheries

Components of the dam

- Heel: contact with the ground on the upstream side
- Toe: contact on the downstream side
- Abutment: Sides of the valley on which the structure of the dam rest
- Galleries: small rooms like structure left within the dam for checking operations.
- Diversion tunnel: Tunnels are constructed for diverting water before the construction of dam. This helps in keeping the river bed dry.
- Spillways: It is the arrangement near the top to release the excess water of the reservoir to downstream side
- Sluice way: An opening in the dam near the ground level, which is used to clear the silt accumulation in the reservoir side.



TYPES OF DAMS

Gravity Dams:

- These dams are heavy and massive wall-like structures of concrete in which the whole weight acts vertically downwards
- As the entire load is transmitted on the small area of foundation, such dams are constructed where rocks are competent and stable.
- Bhakra Dam is the highest Concrete Gravity dam in Asia and Second Highest in the world.
- Bhakra Dam is across river Sutlej in Himachal Pradesh
- The construction of this project was started in the year 1948 and was completed in 1963.
- It is 740 ft. high above the deepest foundation as straight concrete dam being more than three times the height of Qutab Minar.
- Length at top 518.16 m (1700 feet); Width at base 190.5 m (625 feet), and at the top is 9.14 m (30 feet)
- *Bhakra Dam is the highest Concrete Gravity dam in Asia and Second Highest in the world.*

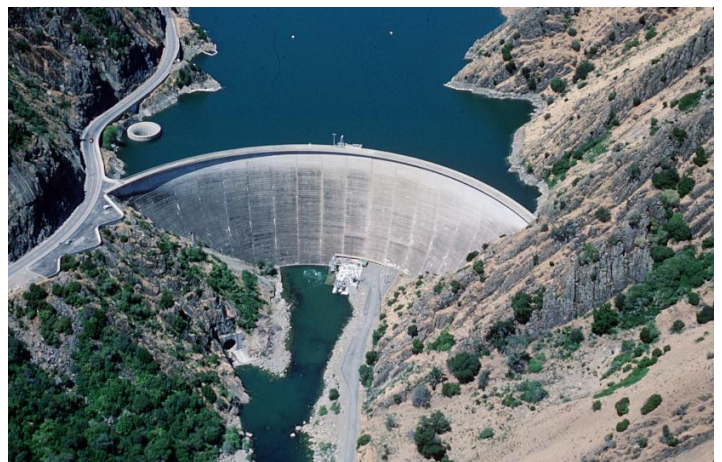


Buttress Dam:

- Buttress Dam – Is a gravity dam reinforced by structural supports like Buttress.
- Buttress is a support that transmits a force from a roof or wall to another supporting structure
- This type of structure can be considered even if the foundation rocks are little weaker

Arch Dam

- These types of dams are concrete or masonry dams which are curved or convex upstream in plan
- This shape helps to transmit the major part of the water load to the abutments
- Arch dams are built across narrow, deep river gorges, but now in recent years they have been considered even for little wider valleys.



Earthen Dams

- They are trapezoidal in shape
- Earth dams are constructed where the foundation or the underlying material or rocks are weak to support the masonry dam or where the suitable competent rocks are at greater depth.
- Earthen dams are relatively smaller in height and broad at the base
- They are mainly built with clay, sand and gravel, hence they are also known as Earth fill dam or Rock fill dam

9. Practice Quiz

1. Which of the following is a primary objective of surveying?

A) Designing buildings

B) Measuring land areas accurately

C) Constructing roads

D) Testing soil samples

2. The instrument used to measure horizontal and vertical angles in surveying is called:

A) Compass

B) Dumpy Level

C) Theodolite

D) Tape

3. Which mode of transportation is best suited for carrying bulk goods over long distances?

A) Roadways

B) Railways

C) Airways

D) Waterways

4. What type of pavement uses layers of bituminous material and granular base?

A) Concrete Pavement

B) Rigid Pavement

C) Flexible Pavement

D) Composite Pavement

5. A red traffic signal indicates:

A) Go

B) Slow down

C) Stop

D) Turn right

6. Which of the following is a correct sequence in the hydrological cycle?

A) Evaporation → Precipitation → Infiltration → Runoff

B) Precipitation → Infiltration → Evaporation → Runoff

C) Runoff → Infiltration → Precipitation → Evaporation

D) Infiltration → Runoff → Precipitation → Evaporation



7. The main source of freshwater is:

- A) Oceans
- B) Lakes and Rivers**
- C) Underground rocks
- D) Glaciers

8. The quality of drinking water is primarily measured by:

- A) Color and taste
- B) pH value, turbidity, and bacterial content**
- C) Smell
- D) Temperature

9. Which structure stores surface runoff for future use and helps recharge groundwater?

- A) Drainage canal
- B) Water treatment plant
- C) Rainwater harvesting system**
- D) Water meter

10. A gravity dam is constructed primarily using:

- A) Steel
- B) Timber
- C) Earth
- D) Concrete or Masonry**

**10. Assignments**

S . No	Question	BL	CO
1	a) What is meant by surveying? and state the objectives of surveying. b) What are the principles of surveying? Explain briefly.	2	3
2	Explain the concept of flexible pavement and mention the advantages and disadvantages of flexible pavement.	2	3
3	Explain the hydrological cycle with neat sketch.	2	3
4	Explain about water storage and water conveyance structures.	2	3

11. Part A- Question & Answers

S.No	Question & Answers	BL	CO
1	Define Surveying. Ans. Surveying is the science and art of determining the relative positions of points on, above, or below the Earth's surface.	1	3
2	State the main principles of surveying. Ans. 1. Working from whole to part 2. Location of a point by measurement from two reference points	1	3

3	What is the object of surveying? Ans. The main object of surveying is to prepare maps and plans for designing and executing civil engineering projects like roads, buildings, and canals.	1	3
4	Differentiate between a plan and a map. Ans. Plan: Drawn to a large scale; used for small areas. Map: Drawn to a small scale; used for large areas.	2	3
5	What are the different modes of transportation? Ans. Roadways Railways Airways Waterways	1	3
6	What is meant by pavement? Ans. Pavement is the top layer of a road structure that provides a smooth and durable surface for vehicles.	1	3
7	List the types of pavements. Ans. Flexible Pavement Rigid Pavement Composite Pavement	1	3
8	Define hydrology. Ans. Hydrology is the science that deals with the study of water in the atmosphere and on the Earth's surface, including its movement, distribution, and quality.	1	3
9	What are the different sources of water? Ans. Rainwater Surface water (rivers, lakes) Groundwater (wells, springs) Desalinated water Recycled water	1	3
10	List the different types of dams. Ans. Gravity Dam Arch Dam Buttress Dam Earth Dam Rock-fill Dam	1	3

12. Part B- Questions

S. No	Question	BL	CO
1	What are the different instruments used in surveying? Explain briefly.	2	3
2	a) Write a note on traffic signs. b) Write a note on traffic signals.	2	3
3	Explain about rain water harvesting and mention the advantages and disadvantages of rain water harvesting.	2	3
4	Explain about water storage and water conveyance structures.	3	3
5	Sketch and explain different types of dams.	3	3

13. Supportive Online Certification Courses

1. A NPTEL Course on Introduction to Civil Engineering Profession

14. Prescribed Text Books & Reference Books**Text Book**

1. Basic Civil Engineering, M.S. Palanisamy, Tata Mc Graw Hill publications (India) Pvt., Ltd., Fourth Edition.
2. Introduction to Civil Engineering, S.S. Bhavikatti, New Age International Publishers, 2022, First Edition.
3. Basic Civil Engineering, Satheesh Gopi, Pearson Publications, 2009, First Edition.

References:

1. Surveying, Vol – I and Vol - II, S.K.Duggal, Tata McGraw Hill Publishers 2019, Fifth Edition.
2. Hydrology and Water Resources Engineering, Santosh Kumar Garg, Khanna Publishers, Delhi, 2016.
3. Irrigation Engineering and Hydraulic Structures –Santosh Kumar Garg, Khanna Publishers, Delhi 2023, 38th Edition.
4. Highway Engineering, S.K.Khanna, C.E.G. Justo and Veeraraghavan, Nemchand and Brothers Publications 2019. 10th Edition.
5. Indian Standard DRINKING WATER — SPECIFICATION IS 10500-2012.

